**QP CODE 25900510** **SEMESTER 3**

# MG UNIVERSITY MGU-UGP (HONOURS) EXAMINATION OCT 2025

## MG3DSCMAT204 – ESSENTIALS OF FINANCIAL MATHEMATICS

### Part A

**(Any 5 questions. Maximum 2 marks can be earned from each question)**

1. Correct Definition or General form ……2 Marks
   (Give 1 mark for any suitable example)
2. Correct Definition ……..……2 Marks
3. Writing R(x)=C(x) ……1 Mark
   Getting the answer 104 ……1 Mark
4. Correct Definition ……2 Marks
5. $9x + 2y + s = 86$ ……1 Mark
   (Any alphabet or symbol can be used in the place of s as the variable)
   Slack Variable ……1 Mark
6. For writing the objective function ……1 Mark
   For writing the constraints ……..……1 Mark
7. Finding $MC = 3Q^2 - 6Q$ ……1 Mark
   Finding it's derivative $MC^{'} = 6Q - 6$ ……1 Mark
   For getting the correct answer $Q = 1$ ……1 Mark
8. Finding $f^{'}(x) = 6x^2 + 18x + 12$ ……1 Mark
   For getting the answers $x = -2$ and $x = -1$ …..……1 Mark

### Part B

**(Any 5 questions. Maximum 6 marks can be earned from each question)**

9. Formula $FV = \frac{(1+r)^n - 1}{r} \times 100$ ……2 Mark
   For substituting $FV = \frac{(1+0.01)^n - 1}{0.01} \times 100$ ……2 Mark
   Steps for finding the value of n ……4 Marks
   Final answer 27 months ……..……1 Mark
10. Formula $A = P(1 + r)^n$ ……1 Mark
    For substituting $A = 1000(1 + 0.08)^n$ ……1 Mark
    Rest ……..……3 Marks
    Final answer 2158.92 (or approximate value)……1 Mark
11. Substituting for C and I in Y= C+I ……1 Mark
    For simplification ……2 Mark

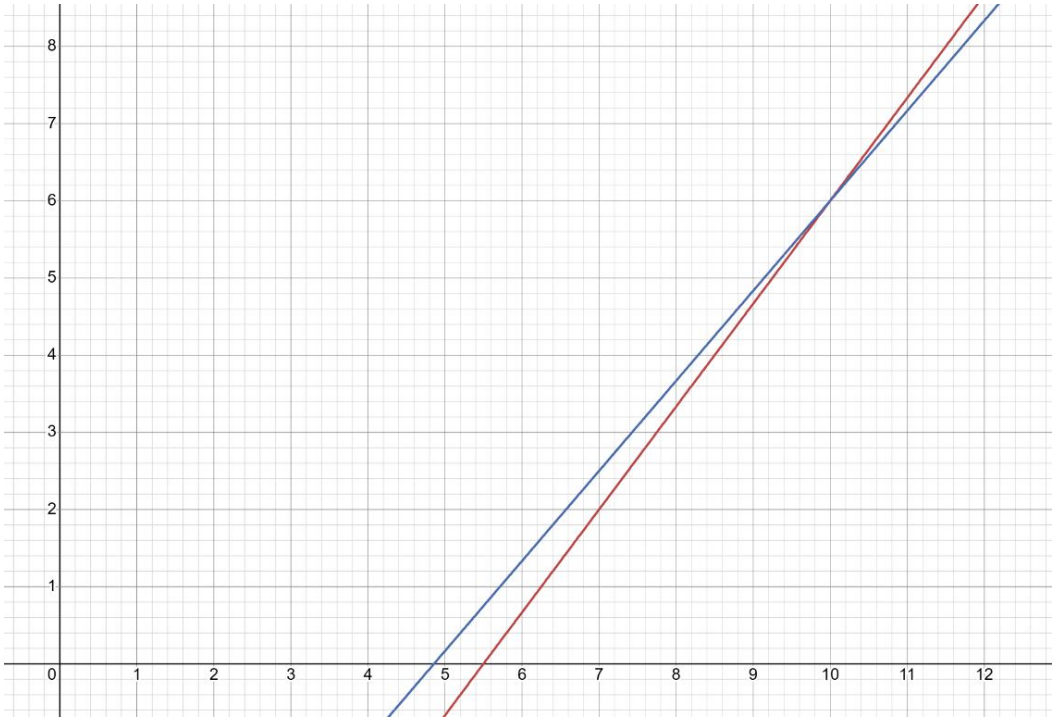
Finding the reduced form $Y = \frac{C_0+I_0}{1-a-b}$ .....1 Mark

For substitution .....1 Mark

For getting the answer Y=750 .....1 Mark

12. For graphing the line $4x - 3y = 22$ .....2 Marks

For graphing the line $7x - 6y = 34$ .....2 Marks

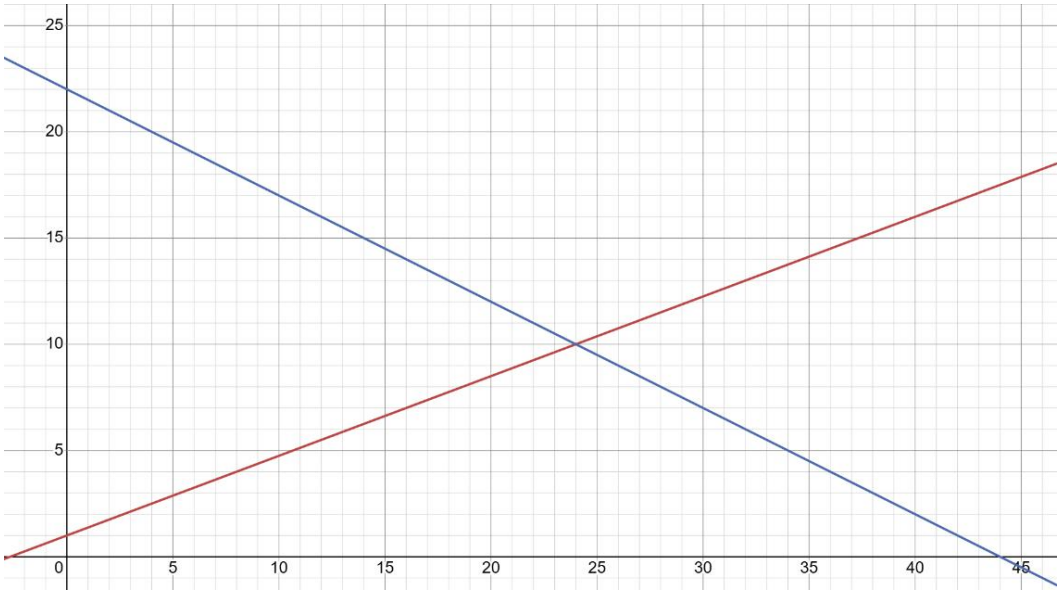



For finding the solution $x = 10$ & $y = 6$ .....4 Marks

13. For finding the equilibrium price $P = 10$ .....3 Marks

For finding the equilibrium quantity $Q = 24$ .....3 Marks

For the graph .....4 Marks



14. For graphing the line $7x + 8y = 168$ .....2 Marks

For graphing the line $2x + 4y = 60$ .....2 Marks




For shading the feasible region .....2 Marks

15. For writing $MC = \frac{d}{dQ}TC$ .....1 Marks

For finding $MC = 0.5 - 2Q$ .....2 Marks

For writing $AC = \frac{TC}{Q}$ .....1 Marks

For finding $AC = 0.5 - Q$ .....2 Marks

16. For finding $\pi' = -6Q^2 - 30Q + 3000$ .....2 Marks

For finding the critical points 20 & -25 .....2 Marks

For finding the second derivative $\pi'' = -12Q - 30$ .....2 Marks

For evaluating $\pi''$ at the critical points and finding the maximum .....2 Marks

## Part C

**(Any 3 questions. Maximum 10 marks can be earned from each question)**

17.(a) $P_0$ be the original value of the car. $P_1=P_0(1-.40)=.60P_0$ .... 3 Marks

$P_2=.70P_1$ ..... 2Marks

$P_2=14700$, $P_1=21000$ ..... 2 Marks

$P_0=35000$, For final answer 35000 ..... 2 Marks

(b) Paid amount =(.25)14700=3675 ..... 3Marks

The value after n years $V_n=14700(0.8)^n$ …… 3Marks

For final answer 9 years …… 2Marks

18.(a)Elimination method …… 4Marks

Finding the solution $x = 10$ and $y = 6$ …… 2Marks

(b) substitution method …… 4Marks

Final Solution …… 2Marks

19.For writing Y=C+I …… 2Marks

For substituting and getting 0.4 Y+150i=209 …… 3Marks

For writing Ms=Mw …… 2Marks

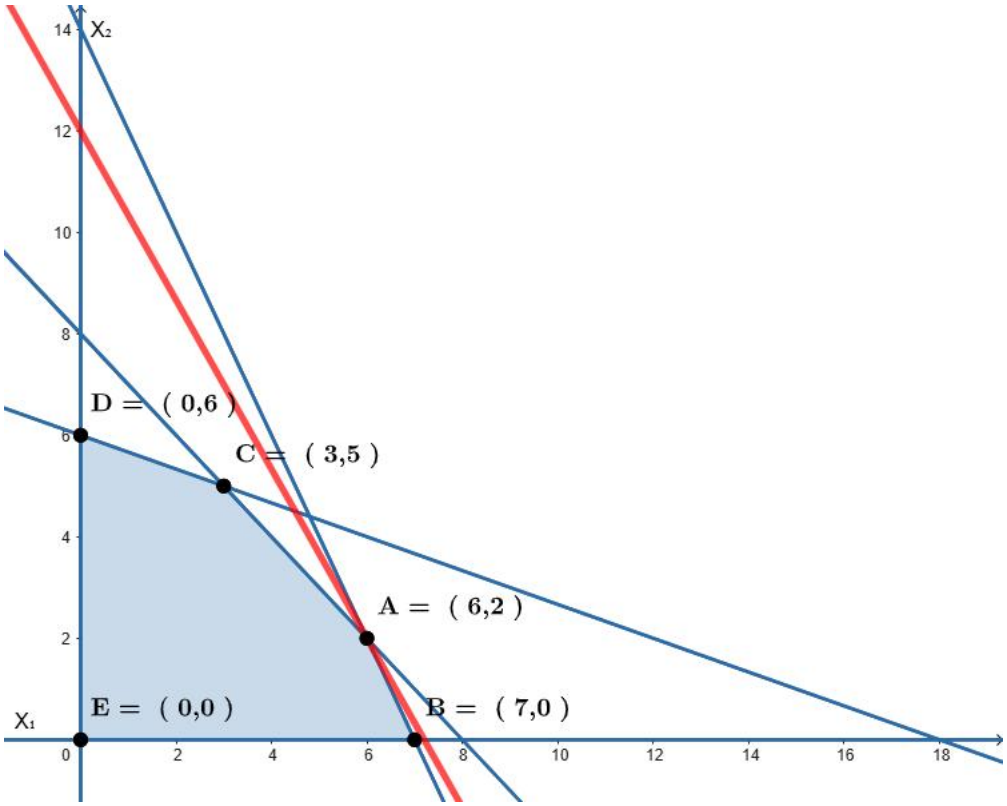
For substituting and getting 0.1 Y-250i=35 …… 3Marks

For solving the system of equations …… 2Marks

For the final Answer Y=500 and $i = 0.06$ …… 2Marks

20.Graphing the 3 lines ……4 Marks

For finding the feasible region ……4 Marks




For finding the optimal solution $x_1 = 6$ and $x_2 = 2$ ……3 Marks

Maximum $z = 360$ ……1 Mark

21. For finding the first derivative ……2 Marks

For finding the critical point $x = 4$ ……3 Marks

For finding the second derivative ……2 Marks

For getting the value of $y''$ equals 0 at the critical point ……..3 Mark

For finding the third derivative ……1 Marks

Rest ……2 Marks

22. For finding the first derivative ……2 Marks

For finding the critical point $x = 6$ ……3 Marks

For finding the second derivative ……2 Marks

For getting the value of *y"* equals 0 at the critical point ……..3 Mark

For finding the third derivative ……1 Marks

Rest ……2 Marks

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